Synoptic Question 9: The Physics of an Offshore Wind Turbine

Reading Passage

Read the following article and then answer the questions that follow.

Wind power is a rapidly growing sector of renewable energy. Modern wind turbines capture the kinetic energy of moving air and convert it into electrical energy. A turbine consists of large blades (typically three) mounted on a rotor, which is connected via a drive shaft and gearbox to a generator. This entire assembly is housed in a structure called the nacelle, which sits on top of a tall tower.

The power available from the wind is proportional to the cube of the wind speed. The kinetic energy of the air passing through the circular area swept by the blades is converted into rotational kinetic energy, which then drives the generator. The theoretical maximum efficiency of any wind turbine is known as the Betz limit, which states that no turbine can capture more than 59.3% of the wind's kinetic energy.

Offshore wind turbines are particularly effective as wind speeds over the sea are generally higher and more consistent than on land. These are some of the largest machines on Earth, with blade lengths exceeding 90 metres. The large rotational speed of the blade tips can be a source of structural stress and significant low-frequency noise.

The electricity generated in the nacelle is at a relatively low voltage. To transport this power efficiently to the shore, it is passed through a transformer to increase the voltage and, consequently, decrease the current, thereby minimizing power loss in the long undersea cables.

Data:

Data for a large offshore wind turbine:

Table 1: Data for Offshore Wind Turbine

Parameter	**Value**
Length of one blade, L	90.0 m
Density of air, ρ	1.25 kg m^{-3}
Operating wind speed, v	12.0 m s^{-1}
Rotational speed of blades, ω_{rpm}	10.0 rpm
Actual electrical power output, P_{elec}	5.00 MW
Efficiency of generator, η_{gen}	94%
Output voltage of generator, V_{gen}	690 V
Total resistance of generator coils, R	0.0050 Ω
Frequency of sound from blades, f	25 Hz
Speed of sound in air, v_{sound}	340 m s^{-1}
Betz's limit, η_{max}	0.593

Questions

(a) Calculate the total area swept by the turbine blades as they rotate.

(b) The power available in the wind, P_{wind}, is given by the formula $P_{wind} = \frac{1}{2}\rho A v^3$.

  (i) Calculate the total power available in the wind passing through the swept area.

  (ii) Calculate the theoretical maximum power, P_{Betz}, that this turbine could extract from the wind.

(c) Using the actual electrical power output, P_{elec}, calculate the overall efficiency of the wind turbine at this wind speed.

(d) The blades rotate at 10.0 rpm (revolutions per minute).

  (i) Convert this rotational speed into angular velocity, ω, in rad s^{-1}.

  (ii) Calculate the linear speed of the tip of a blade.

  (iii) Suggest one reason why it is important to limit this tip speed.

(e) The generator's efficiency is 94%. Calculate the mechanical power, P_{mech}, that the blades and drive shaft supply to the generator.

(f) The mechanical power P_{mech} is related to the torque τ on the drive shaft and the angular velocity ω by the equation $P = \tau\omega$. Calculate the torque exerted by the blades.

(g) Inside the generator:

  (i) Briefly describe how the rotational motion of the drive shaft leads to the generation of an alternating e.m.f.

  (ii) Using the actual electrical power output and the generator voltage, calculate the current I produced by the generator.

  (iii) Calculate the power lost as heat in the generator's resistive coils.

(h) The turbine blades produce a low-frequency sound. Calculate the wavelength of the 25 Hz sound wave.

Solutions

(a) Area $A = \pi L^2 = \pi(90.0)^2 = 25447 \text{ m}^2 \approx 25400 \text{ m}^2$.

(b) (i) $P_{wind} = \frac{1}{2}\rho A v^3 = \frac{1}{2}(1.25)(25447)(12.0)^3$
$P_{wind} = 2.748 \times 10^7 \text{ W} = 27.5 \text{ MW}$.

(ii) $P_{Betz} = P_{wind} \times \eta_{max} = (2.748 \times 10^7) \times 0.593 = 1.63 \times 10^7 \text{ W} = 16.3 \text{ MW}$.

(c) $\eta_{overall} = \frac{P_{elec}}{P_{wind}} = \frac{5.00\times10^6}{2.748\times10^7} = 0.182 = 18.2\%$.

(d) (i) $\omega = 10.0 \frac{\text{rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 1.047 \text{ rad s}^{-1}$.

(ii) Linear speed $v_{tip} = L\omega = (90.0)(1.047) = 94.2 \text{ m s}^{-1}$.

(iii) To prevent structural failure from excessive centripetal forces (stress) or to keep the tip speed below the speed of sound to minimize noise pollution and shock waves.

(e) $\eta_{gen} = \frac{P_{elec}}{P_{mech}}$
$P_{mech} = \frac{P_{elec}}{\eta_{gen}} = \frac{5.00\times10^6}{0.94} = 5.319 \times 10^6 \text{ W} = 5.32 \text{ MW}$.

(f) $\tau = \frac{P_{mech}}{\omega} = \frac{5.319\times10^6}{1.047} = 5.08 \times 10^6 \text{ Nm}$.

(g) (i) The drive shaft spins magnets inside coils of wire (or vice versa). This causes a continuous change in the magnetic flux linkage through the coils. By Faraday's Law of Induction, this changing flux induces an e.m.f. The rotation naturally produces an alternating (sinusoidal) e.m.f.

(ii) $P_{elec} = VI$
$I = \frac{P_{elec}}{V_{gen}} = \frac{5.00\times10^6}{690} = 7246 \text{ A} \approx 7250 \text{ A}$.

(iii) Power loss $P_{loss} = I^2 R = (7246)^2(0.0050) = 262500 \text{ W} = 263 \text{ kW}$.

(h) $\lambda = \frac{v_{sound}}{f} = \frac{340}{25} = 13.6 \text{ m}$.

[Diagram of an offshore wind turbine showing blades, nacelle, and tower]

Figure 1: Offshore Wind Turbine

Cambridge International AS & A Level Physics (9702)

Paper 4 Structured Questions

Variant 2 (Higher Difficulty)

Topic: Offshore Wind Turbine Physics

Instructions

- Answer all questions.
- Show your working and include units.
- For open-ended items, justify your physics reasoning clearly.

Reading Passage (for data extraction)

An offshore wind farm uses turbines with blade length of 95.0 m. At a steady wind speed of $13.5\,\mathrm{m\,s^{-1}}$, the turbine rotor turns at 11.5 rpm. Under these conditions, the nacelle reports an electrical output of 6.20 MW. The generator operates at an efficiency of 0.955 and sends power to shore at 3.30 kV through undersea cables.

During operation, technicians monitor both aerodynamic and acoustic limits. Air density at the site is taken as $1.23\,\mathrm{kg\,m^{-3}}$, while the dominant blade-passing sound has frequency 22.0 Hz. For comparison with physical limits, use Betz limit $\eta_{\mathrm{Betz}} = 0.593$ and the speed of sound in air $c = 340\,\mathrm{m\,s^{-1}}$.

Additional given data

Parameter	Value
Cable resistance to shore (total loop), R_{cable}	$1.80\,\Omega$

Question 1 [30]

(a) Calculate the swept area of the rotor. [2]

(b) The wind power crossing the rotor disk is $P_{\mathrm{wind}} = \frac{1}{2}\rho A v^3$.

(i) Calculate P_{wind}. [2]

(ii) Calculate the maximum extractable power using Betz's limit. [2]

(iii) Determine the turbine's overall aerodynamic-to-electric efficiency at this operating point. [2]

(c) Convert the rotational speed to angular speed in $\mathrm{rad\,s^{-1}}$, then find blade-tip speed. State whether the tip speed is subsonic and determine the Mach number. [4]

(d) The mechanical input power to the generator is related to output by $\eta_{\mathrm{gen}} = P_{\mathrm{elec}}/P_{\mathrm{mech}}$.

(i) Calculate P_{mech}. [2]

(ii) Using $P = \tau\omega$, calculate the shaft torque. [2]

(e) The turbine exports power at 3.30 kV through undersea cables.

(i) Calculate transmission current. [2]

(ii) Calculate cable power loss $P_{\mathrm{loss}} = I^2 R_{\mathrm{cable}}$. [2]

(iii) Calculate transmission efficiency from nacelle electrical output to shore-end cable output. [2]

(f) The dominant sound frequency is 22.0 Hz. Calculate wavelength and comment on whether diffraction around support structures is expected to be significant. [2]

(g) *Open-ended (physics explanation)*: Discuss how and why available wind power and mechanical loading on the turbine change when wind speed increases by 20% from the value in the table, assuming constant air density and unchanged rotor radius. Your answer should include quantitative scaling and at least two engineering consequences. [8]

Sample Solution – Variant 2

1(a)

$$A = \pi L^2 = \pi(95.0)^2 = 2.835 \times 10^4\,\mathrm{m}^2$$

1(b)(i)

$$P_{\text{wind}} = \frac{1}{2}(1.23)(2.835 \times 10^4)(13.5)^3 = 4.29 \times 10^7\,\mathrm{W} = 42.9\,\mathrm{MW}$$

1(b)(ii)

$$P_{\text{Betz}} = 0.593\,P_{\text{wind}} = 0.593(42.9\,\mathrm{MW}) = 25.4\,\mathrm{MW}$$

1(b)(iii)

$$\eta_{\text{overall}} = \frac{P_{\text{elec}}}{P_{\text{wind}}} = \frac{6.20}{42.9} = 0.145 \approx 14.5\%$$

1(c)

$$\omega = 11.5 \times \frac{2\pi}{60} = 1.20\,\mathrm{rad\ s^{-1}}$$

$$v_{\text{tip}} = \omega L = (1.20)(95.0) = 1.14 \times 10^2\,\mathrm{m\ s^{-1}}$$

$$\text{Mach} = \frac{v_{\text{tip}}}{c} = \frac{114}{340} = 0.34$$

Tip speed is subsonic.

1(d)(i)

$$P_{\text{mech}} = \frac{P_{\text{elec}}}{\eta_{\text{gen}}} = \frac{6.20 \times 10^6}{0.955} = 6.49 \times 10^6\,\mathrm{W}$$

1(d)(ii)

$$\tau = \frac{P_{\text{mech}}}{\omega} = \frac{6.49 \times 10^6}{1.20} = 5.40 \times 10^6\,\mathrm{N\ m}$$

1(e)(i)

$$I = \frac{P_{\text{elec}}}{V} = \frac{6.20 \times 10^6}{3.30 \times 10^3} = 1.88 \times 10^3\,\mathrm{A}$$

1(e)(ii)

$$P_{\text{loss}} = I^2 R = (1.88 \times 10^3)^2(1.80) = 6.36 \times 10^6\,\mathrm{W}$$

1(e)(iii)

$$\eta_{\text{trans}} = \frac{P_{\text{elec}} - P_{\text{loss}}}{P_{\text{elec}}} = \frac{6.20 - 6.36}{6.20} \approx -0.026$$

This is non-physical (negative delivered power), indicating the given voltage is too low for this cable resistance and power level; in practice, much higher transmission voltage is required.

1(f)

$$\lambda = \frac{c}{f} = \frac{340}{22.0} = 15.5\,\mathrm{m}$$

Since wavelength is large compared with many structural features, diffraction is significant.

1(g) Open-ended response (sample)

If wind speed increases by 20%, then $v' = 1.20v$.

Available wind power scales as $P_{\text{wind}} \propto v^3$, so

$$\frac{P'_{\text{wind}}}{P_{\text{wind}}} = (1.20)^3 = 1.728$$

This is a 72.8% increase in available power.

Aerodynamic force loading on blades approximately scales as dynamic pressure, so $F \propto v^2$:

$$\frac{F'}{F} = (1.20)^2 = 1.44$$

So thrust and bending loads increase by about 44%, increasing fatigue and peak stress.
Engineering consequences include:

- greater torque and electrical output potential, but only up to generator and converter ratings;
- increased blade-root bending moment, requiring pitch control and structural safety margins;
- higher risk of overspeed, so active braking/pitch-to-feather control is needed;
- potential increase in aeroacoustic noise due to higher tip speed and turbulence.

Mark Scheme – Variant 2 (CIE P4 style)

Question 1

(a) [2]

- B1: Uses $A = \pi L^2$.
- A1: $A = 2.84 \times 10^4\,\mathrm{m}^2$ (allow 2.83–2.84 $\times 10^4$).

(b)(i) [2]

- M1: Substitution into $P = \frac{1}{2}\rho A v^3$.
- A1: $P_{\mathrm{wind}} = 4.29 \times 10^7\,\mathrm{W}$ (42.9 MW).

(b)(ii) [2]

- M1: Uses Betz factor 0.593.
- A1: $P_{\mathrm{Betz}} \approx 2.54 \times 10^7\,\mathrm{W}$ (25.4 MW).

(b)(iii) [2]

- M1: $\eta = P_{\mathrm{elec}}/P_{\mathrm{wind}}$.
- A1: $\eta \approx 0.145$ or 14.5%.

(c) [4]

- B1: Correct rpm to rad s^{-1} conversion method.
- A1: $\omega \approx 1.20\,\mathrm{rad\ s^{-1}}$.
- M1: Uses $v_{\mathrm{tip}} = \omega L$ and/or Mach $= v/c$.
- A1: $v_{\mathrm{tip}} \approx 114\,\mathrm{m\ s^{-1}}$ and Mach ≈ 0.34 (subsonic).

(d)(i) [2]

- M1: Rearrangement $P_{\mathrm{mech}} = P_{\mathrm{elec}}/\eta_{\mathrm{gen}}$.
- A1: $P_{\mathrm{mech}} \approx 6.49 \times 10^6\,\mathrm{W}$.

(d)(ii) [2]

- M1: Uses $\tau = P/\omega$.
- A1: $\tau \approx 5.40 \times 10^6\,\mathrm{N\ m}$.

(e)(i) [2]

- M1: Uses $I = P/V$.
- A1: $I \approx 1.88 \times 10^3\,\mathrm{A}$.

(e)(ii) [2]

- M1: Uses $P_{\mathrm{loss}} = I^2 R$.
- A1: $P_{\mathrm{loss}} \approx 6.36 \times 10^6\,\mathrm{W}$.

(e)(iii) [2]

- M1: Uses efficiency comparison with delivered power.
- A1: Concludes non-physical/very poor transmission at this voltage; states need for higher step-up voltage (or equivalent valid efficiency calculation showing impossible outcome).

(f) [2]

- M1: Uses $\lambda = c/f$.
- A1: $\lambda \approx 15.5\,\mathrm{m}$ and valid diffraction comment.

(g) [8] Open-ended levels-based

- B2: States correct scalings $P \propto v^3$, loading/thrust $\propto v^2$.
- C2: Quantifies changes for +20% wind ($\times 1.728$ power, $\times 1.44$ load).
- C2: Links scaling to turbine operation (torque, fatigue, control limits, cut-out).
- Q2: Coherent, well-structured scientific argument with at least two relevant engineering consequences.

Total: 30 marks